

Motability Operations Delivered a Successful Pilot for Predictive Maintenance Using AWS

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Architect, Motability Operations



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[Motability Operations](#) provides vehicles to over 700,000 disabled people in the United Kingdom through the Motability Scheme. To provide excellent service to its customers, Motability Operations wanted to reduce the number of breakdown incidents by using



connected vehicle data. Unexpected vehicle breakdown is always inconvenient, but for Motability Operations' customers, it can be significantly more challenging. So, the company used Amazon Web Services (AWS) and collaborated with [AWS Professional Services](#)—a global team of experts—to deliver a 6-month pilot for predictive maintenance.

Enhancing Customer Experience with Connected Vehicle Data

To qualify for the Motability Scheme, a customer must receive an eligible mobility allowance. Since 1978, every vehicle lease has breakdown and maintenance included. Using its connected vehicle data, Motability Operations wanted to run a pilot for predictive vehicle maintenance using artificial intelligence and machine learning (AI/ML) services on AWS to enhance the accuracy of its predictions. It recognized that preventing unplanned events, such as vehicle breakdowns, could greatly secure and simplify its customers' driving experience.

Motability Operations worked closely alongside AWS Professional Services to establish an initial proof of concept for its pilot program as it built its technical team. Motability Operations chose to use a variety of AWS serverless and managed services for flexibility, scalability, and cost-efficiency. "Using AWS services was the right decision," says Vijay Srinivasan, principal architect at Motability Operations. "Not worrying about the complexity of infrastructure and complicated security management, we could focus on building architecture that would deliver business value quickly." Motability Operations spent 6 months collecting and training data for its solution's foundational ML models.

Delivering a Successful Pilot Using Amazon SageMaker for ML

Motability Operations, alongside AWS Professional Services, built ML models to determine whether a vehicle was likely to break down in the near future. To create these models, Motability Operations used Amazon SageMaker, which is designed to build, train, and deploy models for virtually any use case with fully managed infrastructure, tools, and workflows.

The pilot consisted of 6,500 vehicles. In total, 49 percent of breakdowns were predicted, and seven major vehicle faults were corrected. "The trial was more successful than we had imagined," says Felicity Kelly, connected and digital innovation lead at Motability Operations. "We were able to intervene proactively, manage cases, and enhance the customer experience."

Customer response throughout the pilot was overwhelmingly positive. Less than 1 percent of people chose to opt out. By successfully predicting maintenance



Motability Operations improved the overall customer experience, thus adding business value.

Completing the Solution Using AWS

Motability Operations plans to conduct follow-up pilots related to features such as customer notifications, data sharing with dealers, and proactive tire maintenance to complete its solution. From there, Motability Operations intends to use AWS to scale the solution to cover its full fleet.

“The breadth of AWS services made our data acquisition journey much easier,” says Vijay Srinivasan. “Whenever new use cases come up, using AWS gives us the confidence that we can deliver effective solutions quickly.”

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